

Volatility Forecasting: GARCH and Recurrent Models

A controlled walk-forward comparison
with calibration and uncertainty diagnostics

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EXPERIMENT SCOPE

Nine specifications, 3,738 common test dates, February 5, 2010 to December 11, 2024. Six neural models each trained across 178 expanding-window folds. One primary training seed.

Research question

Can recurrent models improve one-day equity-index variance forecasts over parsimonious classical baselines? The comparison separates implementation correctness, forecast positivity, mean calibration, and loss-based performance.

What changed

The rerun repairs daily GARCH state updates, corrects a target inverse mismatch, aligns neural training availability, seeds before model construction, and evaluates train/validation loss at the same restored checkpoint. Positive log-ratio hybrids replace the original unconstrained additive residual specification.

Main finding

GARCH-t has the lowest QLIKE point estimate (-8.4613); GARCH-t has the lowest MSE (1.601e-07). The best neural QLIKE point estimate is Ratio LSTM (-6.6595), higher than GARCH-t (-8.4613).

Across the nine complete prediction series, 0 forecasts are at or below $1e-12$. Positivity does not establish calibration or superiority. The study uses a previously examined historical snapshot, not an untouched prospective holdout.

Source: corrected-v1 aggregate results; cached ^GSPC snapshot; no fresh market-data download.

01 / A Comparable Experiment

Element	Corrected design
Data and target	Cached daily ^GSPC prices, 2003-2024. Target y is squared log return in daily decimal-return variance units.
Classical baselines	Student-t GARCH(1,1); lagged mean of 21 squared returns; EWMA with lambda 0.94.
Neural inputs	21 rows ending at t-1: log return, squared return, absolute return, annualized 21-day rolling volatility. Feature/ratio hybrids also include lagged GARCH variance.
Common availability	Modeling frame begins after the 756-return GARCH warmup. Initial neural training span has 777 rows, yielding 756 sequences; validation has 252 observations.
Walk-forward tests	178 non-overlapping blocks of 21 observations. Training expands; validation and model weights are reselected per fold. The incomplete trailing block is excluded.
Neural settings	8 units; dropout 0.1; Adam 0.001; batch 64; maximum 35 epochs; patience 6; seed 42 plus split ID. CPU execution; fixed-length recurrence unrolled.

Forecast timing

A forecast for date t is issued using information through t-1. Feature and target transform statistics are fitted only on the training sample. The validation sample selects early stopping. Later folds may train on observations that were test data in an earlier fold, once those observations become historical.

Why the evaluation now begins in 2010

The original hybrid runs had very little early training history after GARCH warmup. The rerun requires a full shared neural training span and therefore starts testing later. Old and new loss levels must not be interpreted as a one-change before/after comparison. GARCH retains its own expanding return-estimation history; equal neural dates do not make every estimator's effective memory identical.

Squared returns are a noisy second-moment proxy. Their interpretation as conditional variance additionally assumes a zero conditional mean. The 2008 crisis is in training history, not this corrected test period.

02 / Results on Identical Test Dates

All rows use the same 3,738 targets and dates. Three classical baselines and six freshly trained neural specifications are included. This is a new historical walk-forward run, not a rescore of the old forecasts.

Forecast	MSE / 10 ⁻⁷	QLIKE	Mean h / mean y
GARCH-t	1.60062	-8.46125	1.038
Lagged RV21	1.87208	-8.35289	1.005
EWMA 0.94	1.74818	-8.41999	1.004
Pure LSTM	1.93876	-6.44830	0.258
Pure GRU	1.91925	-6.43031	0.259
Feature LSTM	1.96415	-6.60657	0.262
Feature GRU	1.97011	-6.51491	0.256
Ratio LSTM	1.86316	-6.65949	0.299
Ratio GRU	1.85128	-6.57178	0.283

$$\text{MSE} = \text{mean}[(y - h)^2]$$

$$\text{QLIKE} = \text{mean}[\log(h) + y/h]$$

Both losses are lower-is-better. QLIKE uses a fixed 1e-12 numerical floor for y and h; no floor was selected against test losses. Negative QLIKE levels are normal under this convention. The MSE column is scaled by 10⁻⁷; these are not returns or percentages.

Interpretation

GARCH-t has the lowest QLIKE point estimate (-8.4613); GARCH-t has the lowest MSE (1.601e-07). The best neural QLIKE point estimate is Ratio LSTM (-6.6595), higher than GARCH-t (-8.4613).

Mean h / mean y is a coarse level-calibration diagnostic, not a full conditional calibration test. A value below one means the average variance prediction is below the average realized proxy. A good average ratio does not rule out tail or regime failures.

Both original-scale MSE and QLIKE can be robust to an unbiased noisy proxy under suitable conditions [2]. Different loss weighting and finite-sample behavior can produce different rankings. Point-estimate selection across several candidates does not establish statistically reliable superiority.

03 / Paired Loss Uncertainty

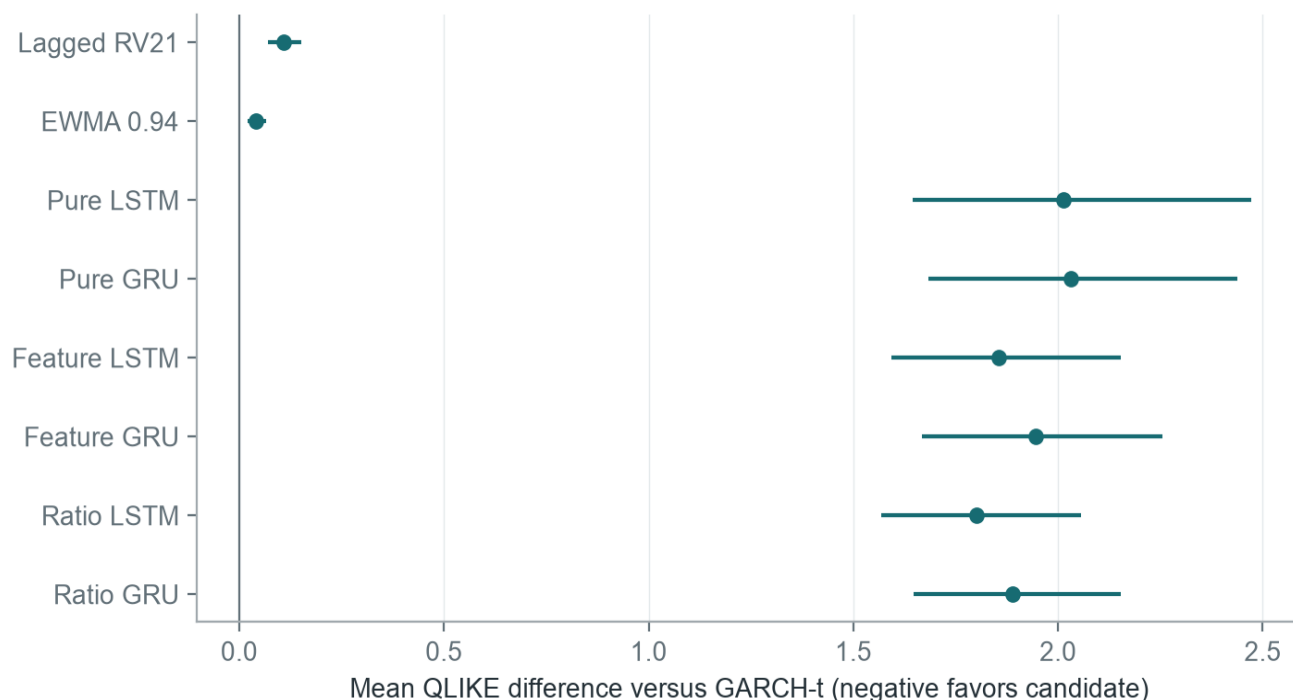


Figure 1. Candidate minus GARCH-t mean QLIKE, with 95% pointwise percentile intervals from a paired stationary bootstrap. Average block length 21; 2,000 replications; seed 20260924. Negative values favor the candidate. All comparisons use 3,738 dates.

What the interval addresses

Daily candidate and benchmark losses are paired by date. Resampling blocks retains some serial dependence rather than treating daily observations as independent. The interval concerns the mean loss difference conditional on these trained forecasts and suitable time-series assumptions.

For Ratio LSTM, the observed QLIKE difference versus GARCH is **+1.8018**, with interval **[+1.5671, +2.0561]**. This model is highlighted because it has the lowest neural QLIKE point estimate; that selection is post hoc and is not accounted for by the individual interval.

What it does not address

These are not simultaneous intervals across all candidates and do not quantify neural initialization uncertainty, data-vendor variation, research selection, or future regime shifts. Block-length sensitivity at 5 and 63 is retained in the aggregate evidence file. A single historical comparison is not proof of deployment value.

04 / Positivity Is Not Calibration

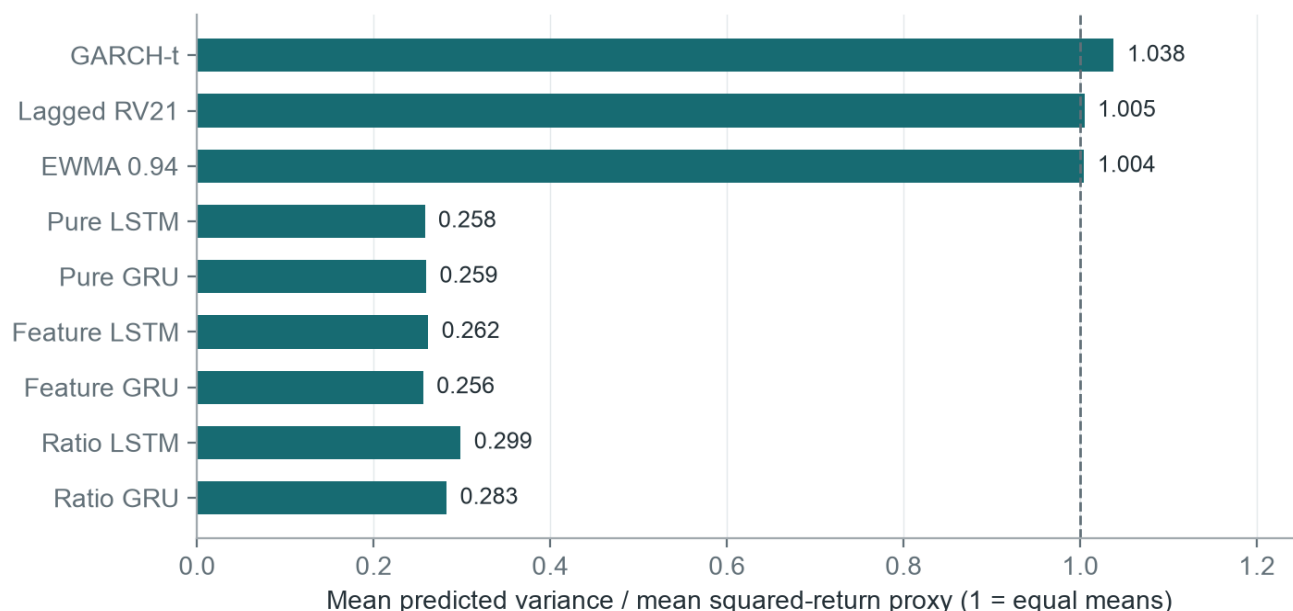


Figure 2. Mean predicted variance divided by mean squared-return proxy on the common test sample. The dashed line marks equal means; it is not a significance threshold.

A positive hybrid construction

The ratio hybrid learns $z = \log(\max(y, 1e-8)/g)$, where g is GARCH variance, and reconstructs $h = g \exp(\text{predicted } z)$. Finite log predictions and positive g yield positive variance; numeric overflow or underflow is rejected. The target floor prevents $\log(0)$, and is distinct from an output floor.

The objective still matters

The pure/feature models train on standardized log targets; ratio models train on standardized log ratios. Squared error on $\log y$ targets $E[\log y \mid \text{information}]$. Exponentiating does not generally recover $E[y \mid \text{information}]$. Correcting the inverse transform therefore fixes a bug, not this estimand mismatch.

Under a conditional Gaussian illustration without clipping, $\exp(E[\log(r^2)])$ is approximately 0.281 times the conditional variance. This explains why well-fitted log-target models can underforecast arithmetic variance. It is not an empirical correction factor for this dataset.

A next controlled study should compare a positive output trained directly under QLIKE or another original-scale objective. No such additional model is claimed as completed here.

05 / Calendar-Period Diagnostics

The following calendar slices were fixed before inspecting the new test losses. They are descriptive partitions, not a fitted regime classification. Each column compares models on the same dates within that period.

Forecast	2010-2016	2017-2019	2020-2021	2022-2024
GARCH-t	-8.490	-8.950	-7.998	-8.211
Lagged RV21	-8.365	-8.816	-7.872	-8.182
EWMA 0.94	-8.458	-8.881	-7.906	-8.213
Pure LSTM	-6.887	-7.632	-4.054	-5.846
Pure GRU	-6.793	-7.664	-4.303	-5.772
Feature LSTM	-6.963	-7.570	-4.837	-5.995
Feature GRU	-6.842	-7.594	-4.894	-5.753
Ratio LSTM	-7.074	-7.307	-5.270	-5.973
Ratio GRU	-6.995	-7.284	-5.118	-5.844

Table 2. Mean QLIKE by period; lower is better. Observation counts: 2010-2016: 1,739; 2017-2019: 754; 2020-2021: 505; 2022-2024: 740.

Why inspect subperiods?

An aggregate mean can hide unstable relative performance or a handful of large loss contributions. Calendar diagnostics help identify questions for follow-up, but choosing favorable periods after seeing them would be a selection problem.

What remains untested

The main run uses one initialization schedule. It does not establish seed robustness, performance across assets, conditional calibration, or stability under a different target proxy. The historical dataset has been studied before; this is not an untouched final holdout.

Trading relevance

Variance forecasts may inform risk sizing, volatility targeting, or risk estimates. Those are potential applications, not results of this study. No strategy, execution model, transaction-cost analysis, portfolio P&L, or Sharpe-ratio improvement is claimed.

The appropriate conclusion is specific to the forecast target, information set, sample, objective, and baselines actually tested. More complex architecture is not itself evidence of better research or better investment performance.

06 / Recurrent Gates as Diagnostics

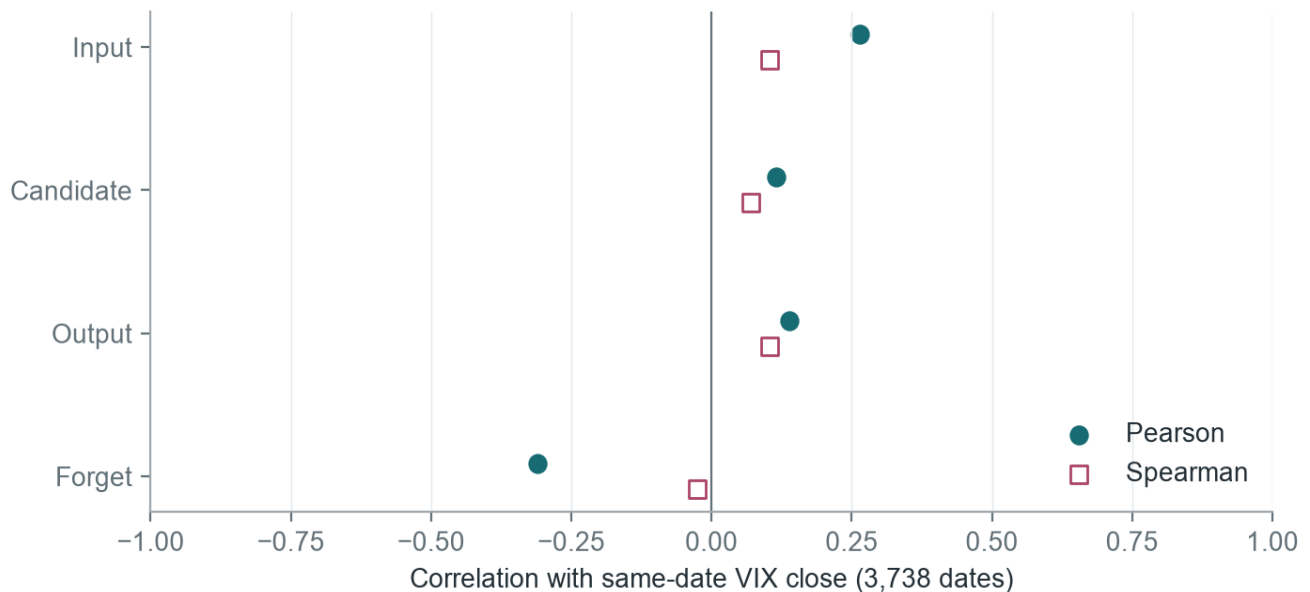


Figure 3. Corrected-run pure-LSTM activations, averaged across units and lags 1-20, versus same-date VIX close. Pearson and rank correlations use 3,738 matched dates. The appendix lists all 21 activation/variant summaries.

What is observable

The pure-LSTM forget-gate aggregate has Pearson correlation **-0.310** and Spearman correlation **-0.025** with VIX in this rerun. These are descriptive associations, not identified mechanisms.

Three interpretation limits

Timing: same-date VIX close is ex-post context and was not a predictor available at the previous-day forecast origin.

Pooling: the series combines separately trained models across folds; unit identities and fitted behavior need not remain stable.

Identification: an average activation does not identify a memory horizon, causal response, or tradable signal.

Stronger evidence would require controlled ablations, within-fold checks, multiple seeds, and correctly lagged predictive tests. Candidate activations are shown with the gate summaries for consistency; they are not sigmoid gates.

07 / Conclusions and Provenance

The corrected run demonstrates a reproducible research workflow with explicit forecast timing, shared neural sample availability, positive hybrid construction, simple baselines, and paired temporal uncertainty. The numerical and timing defects identified in the previous report have regression coverage.

GARCH-t has the lowest QLIKE point estimate (-8.4613); GARCH-t has the lowest MSE ($1.601e-07$). The best neural QLIKE point estimate is Ratio LSTM (-6.6595), higher than GARCH-t (-8.4613).

Next research priorities

First, align a positive variance output with an original-scale proper loss. Second, repeat across seeds and assets with a genuinely held-out evaluation plan. Third, connect forecasts to a prespecified economic objective only if the statistical evidence justifies the added complexity.

References

- [1] Bollerslev (1986), Generalized autoregressive conditional heteroskedasticity. [Journal of Econometrics](#).
- [2] Patton (2011), Volatility forecast comparison using imperfect volatility proxies. [Journal of Econometrics](#).
- [3] Hochreiter and Schmidhuber (1997), Long Short-Term Memory. [Neural Computation](#).
- [4] Cho et al. (2014), Learning Phrase Representations using RNN Encoder-Decoder. [arXiv:1406.1078](#).
- [5] [arch forecasting documentation](#) and [stationary bootstrap implementation](#).

Research record

The public repository includes code, tests, pinned research dependencies, this report builder, and aggregate results with input/code/forecast hashes. Raw market data and complete prediction CSVs remain local. The previous report is preserved in Git history; corrected-v1 is a separate experiment, not a silent overwrite of historical scores.

The underlying project originated in collaborative research. This revised portfolio report is presented by Eddie Liu. Contributor names and course identifiers are omitted; the byline does not imply sole authorship of the original experiments.

github.com/SRG-eddieliu/rnn-volatility-lab

Volatility Forecasting: Diagnostic Appendix

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A / Verification Summary

Check	Observed evidence
Run coverage	6 neural specifications x 178 folds = 1,068 fits. Nine complete forecast series, each with 3,738 identical dates and targets.
GARCH fitting	228 parameter refits; all convergence flags are zero. 0 / 4778 consecutive forecasts repeat exactly.
Forecast validity	All final forecasts are finite and positive; 0 at or below 1e-12 across all nine series.
Regression suite	Tests cover the actual arch fixed-parameter reference, new-return response, forecast timing, transforms, input validity, sequences, bootstrap mechanics, and CPU training/resumption.
Metric verification	Independent array calculations reconcile with the repository MSE and QLIKE functions. All model CSV hashes match their completion records.

Repaired daily recursion

$$h_t = \text{omega} + \text{alpha } r_{t-1}^2 + \text{beta } h_{t-1}$$

On refit dates, parameters and endpoint variance come from an expanding fit using returns through t-1. Between refits, parameters stay fixed while every new return updates the state. Internal percent-squared variance is divided by 10,000 for the output units.

The corrected experiment does not certify upstream price history against an independent vendor, every unused legacy notebook, or production controls. A converged fit is a numerical condition, not proof that the model is correct.

B / Training and Numerical Checks

Neural model	Mean epochs	Train loss	Validation loss	Paired gap
Pure LSTM	27.1	0.8001	0.8252	+0.0251
Pure GRU	25.5	0.8016	0.8247	+0.0231
Feature LSTM	26.2	0.8009	0.8257	+0.0248
Feature GRU	25.2	0.8023	0.8251	+0.0228
Ratio LSTM	27.1	0.9344	0.9681	+0.0337
Ratio GRU	26.3	0.9396	0.9700	+0.0303

Table A1. Means across 178 folds. Train and validation losses are evaluated in inference mode at the same restored best-validation checkpoint. The losses are standardized-target MSE, not original-scale variance MSE.

Interpretation

The corrected paired gap fixes the old practice of subtracting independent minima from different epochs. A gap is still a diagnostic, not a theorem about overfitting. Pure/feature log targets and ratio targets differ, so cross-family gap magnitudes are not directly comparable.

Transform and reconstruction

For log targets, the forward map uses $\log(\max(y, 1e-8))$; the inverse exponentiates without subtracting epsilon. Clipping is not invertible below its floor, and the tests explicitly check this boundary behavior.

For ratio hybrids, the signed log-ratio target is standardized without taking another logarithm. Reconstruction uses $\exp(\log(g) + \text{predicted } z)$ in float64 and rejects non-finite or nonpositive numerical results.

Reproducibility and interruption handling

Seeds are set before model construction. TensorFlow deterministic CPU operations are enabled. A run signature binds input bytes, relevant code, configuration, and split selection. An output directory with a conflicting signature is rejected. A split is complete only when predictions, required gate rows, and the final training log agree on coverage.

Deterministic smoke tests reproduce within the tested environment. This is not a guarantee of bitwise portability to another TensorFlow version or hardware platform.

C / Paired Loss-Difference Intervals

MSE differences / 10^{-7}

Candidate vs GARCH	Difference	95% lower	95% upper
Lagged RV21	+0.27145	+0.00863	+0.78988
EWMA 0.94	+0.14756	-0.02708	+0.47145
Pure LSTM	+0.33813	-0.04210	+1.08428
Pure GRU	+0.31863	-0.06924	+1.04830
Feature LSTM	+0.36353	-0.05193	+1.17966
Feature GRU	+0.36948	-0.02465	+1.16004
Ratio LSTM	+0.26254	-0.06901	+0.86248
Ratio GRU	+0.25065	-0.06330	+0.81592

QLIKE differences

Candidate vs GARCH	Difference	95% lower	95% upper
Lagged RV21	+0.10836	+0.07105	+0.15051
EWMA 0.94	+0.04126	+0.01967	+0.06587
Pure LSTM	+2.01296	+1.64586	+2.47113
Pure GRU	+2.03095	+1.68416	+2.43884
Feature LSTM	+1.85468	+1.59232	+2.15396
Feature GRU	+1.94634	+1.66693	+2.25586
Ratio LSTM	+1.80176	+1.56706	+2.05614
Ratio GRU	+1.88947	+1.64687	+2.15347

Each interval is pointwise, conditional on the trained forecast series, and not adjusted for comparing several candidates. Paired stationary bootstrap; mean block length 21; 2,000 replications. The full aggregate JSON includes 5- and 63-observation block sensitivities with 1,000 replications each. Negative values favor the candidate.

D / Complete Activation Summary

Each row uses the corrected common test period and same-date VIX. Values are averaged across units and lags 1-20, then pooled across rolling folds.

Model	Activation	n	Pearson	Spearman
Pure LSTM	candidate	3738	+0.116	+0.071
Pure LSTM	forget	3738	-0.310	-0.025
Pure LSTM	input	3738	+0.264	+0.104
Pure LSTM	output	3738	+0.139	+0.104
Pure GRU	candidate	3738	+0.185	+0.213
Pure GRU	reset	3738	+0.034	+0.015
Pure GRU	update	3738	+0.417	+0.378
Feature LSTM	candidate	3738	+0.077	+0.075
Feature LSTM	forget	3738	-0.118	+0.054
Feature LSTM	input	3738	+0.288	+0.080
Feature LSTM	output	3738	+0.118	+0.084
Feature GRU	candidate	3738	+0.205	+0.230
Feature GRU	reset	3738	+0.052	+0.049
Feature GRU	update	3738	+0.329	+0.274
Ratio LSTM	candidate	3738	+0.130	+0.097
Ratio LSTM	forget	3738	-0.361	-0.114
Ratio LSTM	input	3738	+0.182	+0.116
Ratio LSTM	output	3738	-0.213	-0.144
Ratio GRU	candidate	3738	+0.231	+0.228
Ratio GRU	reset	3738	+0.025	+0.021
Ratio GRU	update	3738	+0.232	+0.185

These are ex-post correlations without interval estimates or a causal interpretation. Candidate activations are not sigmoid gates. Raw gate outputs, forecasts, and training logs are retained locally for inspection.

E / Reproducibility and Evidence

Package	Recorded version
numpy	1.26.4
pandas	2.2.3
scipy	1.15.3
arch	7.2.0
tensorflow	2.19.0
keras	3.10.0

Rerun commands

Install requirements-research.txt in an isolated environment.

1. Run scripts/run_corrected_experiment.py with --raw, --out, and --prepare.
2. Run scripts/run_all_models.py with the same --out and a bounded --workers value.
3. Run scripts/evaluate_corrected_experiment.py with --out, --destination, and the optional VIX snapshot.
4. Build the PDFs with reports/build_corrected_report.py from the aggregate results JSON.

Artifact identification

The aggregate results record the raw snapshot SHA-256, relevant source-code hashes, configuration, model prediction hashes, and matched VIX snapshot hash. The public repository omits raw market data and prediction CSVs. Hashes identify the local evidence but do not replace access to it for an independent full rerun.

Limits carried into the conclusions

One primary seed; one equity index; a cached historical dataset already examined in earlier research; a noisy daily proxy; log-objective mismatch; pointwise bootstrap uncertainty; no economic strategy or deployment test. The calendar slices and benchmark checks do not remove these limitations.

The underlying project originated in collaborative research. This revised portfolio report is presented by Eddie Liu. Contributor names and course identifiers are omitted; the byline does not imply sole authorship of the original experiments.

Aggregate evidence: reports/corrected_results.json. Source workflow and interpretation boundaries: docs/corrected-experiment-plan.md and docs/validation-status.md.